

Haoyan (Kilian) Shi

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PROFESSIONAL SUMMARY

ML/AI Research Engineer with two peer-reviewed publications (IEEE ISBI 2026, MICCAI 2026) in medical image and video segmentation. Experienced in designing and implementing deep learning architectures spanning Vision Transformers, foundation models, and RAG-based AI systems. Proven ability to contribute from algorithmic design through deployment, with hands-on engineering experience in both research and industry settings.

EDUCATION

Carnegie Mellon University

May 2028

Master of Science in Mobile and IoT Engineering (MSMITE). Expected.

Chongqing University

Jul 2026

Bachelor of Engineering in Computer Science and Technology. TOEFL: 102 | GRE: 331. GPA: 88/100.

PROJECTS

U-KAST: Kolmogorov-Arnold Swin Transformer for Medical Image Segmentation

Aug 2025 – Oct 2025

Researcher, University of Notre Dame

- Accepted by IEEE ISBI 2026 (arxiv.org/abs/2511.04084). Developed U-KAST by integrating Group Rational KANS (GR-KANS) into Swin Transformer encoders to replace standard MLPs, overcoming the computational overhead and memory inefficiencies of vanilla spline-based KANs.
- Parameterized edge functions using Safe Pade Activation Units and channel group-sharing, yielding a highly expressive feed-forward pathway that reduces computational cost (1.4386 GFLOPs) while maintaining a compact footprint (7.2M parameters).
- Alleviated the data-hungry limitations of Vision Transformers, achieving SOTA accuracy across four diverse 2D and 3D medical imaging benchmarks; demonstrated superior robustness and parameter efficiency in data-scarce settings (e.g., using 10% to 25% of training data).

D-GEM: Sparsely Supervised Surgical Video Segmentation

Aug 2025 – Feb 2026

Researcher, University of Notre Dame

- Accepted by MICCAI 2026. Collaborated on the D-GEM framework built upon self-supervised DINOv3 representations, which decouples the memory mechanism from the encoder for architecture-agnostic and data-efficient domain adaptation.
- Contributed to the algorithmic design of the reliability-gated dual-memory system; leveraged deep prior understanding of SAM2/SAM3 temporal propagation to formulate gating heuristics and cosine similarity metrics for memory updates.
- Conducted extensive ablation studies to validate the decoupled memory architecture, systematically evaluating and optimizing configurations for the Gated Transient Memory (GTM) and Evolving Anchor Memory (EAM) to ensure stability under bounded constraints.
- Supported the team in achieving 74.2% average mIoU across three surgical video benchmarks under 10-frame sparse supervision, outperforming SAM3 by 10.3% and mitigating temporal drift on long-horizon sequences.

SIGN: Sperm Head Morphology Classification with Vision Foundation Models

Aug 2025 – Mar 2026

Researcher, University of Notre Dame

- Built an INR-guided framework leveraging SAM3 and frozen Vision Foundation Models (DINOv3, ResNet50), replacing heavy dual-stream convolutional branches to reduce architectural complexity and computational cost.
- Implemented a SIREN-based DeepSDF auto-decoder to compress 2D masks into 256-D latent codes; injected shape priors into backbone features via Feature-wise Linear Modulation (FiLM) structured by contrastive and diversity losses.
- Achieved SOTA performance on two clinical benchmarks, outperforming the SHMC-Net baseline with a 69.1 F1-score on SCIAN-PA and a 98.7 F1-score on HuSHeM.

Research on Mitigation of SAM2 Performance Degradation in Multi-Frame Prompting

Jul 2025 – Dec 2025

Researcher, University of Notre Dame

- Investigated video object segmentation degradation in SAM2 under multi-frame prompting; explored a ControlNet-inspired dynamic gating architecture to integrate multi-frame features.
- Identified limitations of strict temporal propagation and pivoted to an innovative non-sequential propagation strategy.
- Designed and implemented a clustering-based approach to group visually similar frames, routing propagation based on semantic similarity rather than chronological order to mitigate error accumulation empirically.

Two-Stage Intelligent Recognition System for Dental Lesions

Sep 2025 – May 2026

Undergraduate Thesis Project, Chongqing University

- Developed a decoupled two-stage diagnostic pipeline; fine-tuned YOLOv8-Seg to extract fine-grained instance masks and positional priors (0.782 mAP), mitigating cross-tooth noise.
- Enhanced a Swin Transformer backbone by designing a Panoramic Context Module (PANO) and Mask-Guided FiLM to restore global anatomical topology.

- Integrated Rank-MIL pooling with Asymmetric Loss to capture micro-lesion features and resolve long-tail disease distributions.
- Achieved 0.8613 F1 and 0.8610 AUC, outperforming modern baselines (e.g., SwinViT, ConvNeXt); optimized inference to 110 FPS and deployed as a real-time web application.

Abnormal Scene Recognition for Medical Imaging Applications

Nov 2024 – Jan 2025

Researcher, Chongqing University

- Adapted a Vision-Language Model (CLIP + ViT) with Multi-Head Attention to fuse heterogeneous clinical data (EHRS, MRI, real-time feeds) for automated patient anomaly detection and real-time clinical alerts.

MIPS CPU & SoC Design

Dec 2024 – Jan 2025

Developer, Chongqing University

- Designed a 5-stage pipelined MIPS CPU (57 instructions) with a comprehensive hazard unit and data forwarding.
- Constructed a complete SoC by integrating the CPU core with memory via an AXI bus and a write-back cache.

EXPERIENCE

AI Intern

Sep 2025 – Nov 2025

Ericsson (China)

- Developed an end-to-end Fusion-RAG AI search platform deployed via Django, integrating BM25 and ChromaDB for hybrid retrieval with Reciprocal Rank Fusion (RRF) to enhance search relevance.
- Established an automated LLM-as-a-Judge evaluation pipeline (Hit Rate@5/10, MRR) by dynamically generating QA datasets, driving a ~10% overall improvement in retrieval accuracy.
- Optimized context window utilization and latency by decoupling retrieval data and building a HyDE-inspired query reformatting module to map complex user intents.
- Refined system UI/UX and full-stack functionalities, implementing real-time database syncing, dynamic retrieval-count controls, and isolated data access boundaries for enterprise use cases.

Back-End Development Intern

Jun 2025 – Jul 2025

Beijing Silu Innovation Technology

- Migrated a legacy web module from Spring MVC to a modern stack utilizing Spring Boot and Vue.
- Created a generic web scraping tool and automated Docx generation pipelines for Git/SVN repositories.

SKILLS

Frameworks & Tools: PyTorch, MONAI, FastAPI, Django, Spring Boot, Docker, Git

Models & Methods: Swin Transformer, SAM2/SAM3, DINOv3, CLIP, ViT, ResNet50, YOLOv8, KAN, SIREN, ControlNet, BM25, ChromaDB, RAG

Languages: Python, Java, C/C++, SQL